





# Investment cycle of the Brazilian economy: a panel cointegration analysis of industrial firms based on Minsky's financial instability hypothesis—2007–2017

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## ABSTRACT

The aim of this paper is to argue that one of the impacts of the 2008 international financial crisis and the European crisis in the Brazilian economy was in the investment rate. Our interpretation is that the impact of the international crises made the balance sheet of non-financial firms more fragile, mainly the industrial firms, and the economic measures implemented to sustain profit margins from 2011 onwards were not enough to counterbalance the deceleration in the aggregate demand and the rise in financial costs. To investigate the evolution of the financial fragility of industrial firms, following Minsky's financial fragility hypothesis, we propose an econometric model using a cointegration panel. We develop the analysis in two stages using instrumental variables to estimate the rate of investment and the share of financial receipts in total receipts. Among the instrumental variables, the share of wages in value-added, the domestic interest rate, the exchange rate, and the world income were statistically significant. We conclude that the deterioration of the capacity of industrial firms to generate enough internal funds to face debt commitments can be explained by the deceleration of the aggregate demand, expressed by investment rate and the deceleration of the world income, and the increase in the weight of the wage share in total value-added.

## KEYWORDS

Brazilian economy;  
investment cycle; margins  
of safety

## JEL CLASSIFICATIONS

E12; E22; O16

## Introduction

This paper argues that one of the impacts of the 2008 international financial crisis and its unfolding in the years following was in the investment rate. Our interpretation is that the impact of the international crisis made the balance sheet of non-financial firms more fragile, mainly the industrial firms, and the economic measures implemented to sustain profit margins from 2011 onwards were not enough to counterbalance the deceleration in the aggregate demand and the rise in financial costs. Moreover, the 2011–2012 European crisis deepened the world degree of uncertainty,

reaching the domestic economy, affecting the financing strategy of non-financial firms in Brazil. Actually the financial integration of the domestic economy has created conditions for the expansion of private debt issuances, making the economy more sensitive to international events. Therefore, policy space for sustaining aggregate demand has been narrowed in the 2010s.

In order to discuss the building up of the financial fragility in the Brazilian economy, this paper is divided into the following sections. The first briefly discuss Minsky's financial fragility hypothesis, considering the asymmetrical insertion of developing economies in the international financial markets, which tends to reduce their policy space. The second section presents, in the light of the theoretical arguments of the previous section, a discussion about the evolution of the Brazilian economy after the 2008 international financial crisis, emphasizing the behavior of investment rate. It is assumed that the international financial crisis hit the Brazilian economy in an ascending phase of the investment cycle. In order to deep the investigation on the evolution of the financial fragility of industrial firms, the third section presents an econometric exercise to explain its determinants, based on data from the universe of industrial firms in the period 2007–2017. The econometric model uses a cointegration panel. The analysis is developed in two stages using instrumental variables to estimate the rate of investment and the share of financial receipts in total receipts. Among the instrumental variables, the share of wages in value-added, the domestic interest rate, the exchange rate and the foreign income were statistically significant. We conclude that the deterioration of the capacity of industrial firms to generate enough internal funds to face debt commitments can be explained by the deceleration of the aggregate demand, expressed by the investment rate and the deceleration of the world income, and the increase in operational cost through the increase in the weight of the wage share in total value-added. The last section concludes the paper.

### **Minsky's financial instability hypothesis in developing economies and policy space**

The core of Minsky's financial instability hypothesis (1986, 1992) is the interconnection between the investment process and its financing. The financial interaction among economic agents describes how moments of stability and economic turbulence are recurrent.

Under the assumption that decisions are guided by expectations formed under non-probabilistic uncertainty, the endogenous character of the business cycle can be understood as the result of a rational reaction of economic agents in defense of their wealth. Decisions that involve a large amount of financial resources and a long period to mature end up

depending mainly on decisions taken by financial firms to expand financing to meet the demand for capital goods or to reverse the accumulated weight of debt. In times of optimistic expectations and growing aggregate demand, liquidity is plentiful and agents can count on the expected income to cover their debt commitments and on several forms of external financing.<sup>1</sup> However, a deterioration of expectations will point toward a reduction in the supply of new loans and the raising of interest rates

One crucial concept in Minsky's financial instability hypothesis is the erosion of the margins of safety along the business cycle. Minsky (1986, 208) argues that "The margins of safety required by both the borrowers and the lenders affect the extent to which positions and investments are externally financed." Under the financial instability hypothesis, declining margins of safety will be the result of high expectations on future returns. While over-optimistic expectations prevail, balance sheets become more fragile to sudden changes in financial conditions. Kregel (1997) shows that the erosion of the margins of safety is the result of the way that borrowers and lenders evaluate their risks. In his words: "The more things change, the more things appear to remain the same as far as margins of safety are concerned" (548).

The upward phase of the business cycle can be described as the one in which economic units move from more conservative postures in their balance sheets to more speculative postures. While the financial conditions feed positive expectations about the realization of expected profits, liquidity is plentiful. Once financial conditions change because of a change in expectations about future prospects, more indebted units face more difficulty in rolling over their past commitments and the reverse of the expansionary phase is observed.

The interpretation of Minsky's contribution to developing economies, assumed to be dependent on foreign savings, should consider their asymmetrical insertion in the international financial markets.<sup>2</sup> This peculiarity implies that developing economies narrow their policy space if they rely on external savings to sustain growth, as their macroeconomic policies should be oriented to attract foreign capital in the reserve currency.<sup>3</sup> To put this in other words, due to the peculiar way of financial integration, developing economies become more susceptible to a change in the state of expectations of the foreign investors, and because of that the degree of uncertainty driving investment expectations in the domestic economy is higher.

Therefore, capital liberalization in developing economies may not, and in general it does not positively affect the rate of aggregate productive investment. Inversely, capital liberalization in developing economies is associated with the process of financialization, generally defined as a process in which the increasing level of indebtedness of non-financial firms may not be



**Figure 1.** Gross Fixed Capital Formation as a share of GDP (%). Total Economy and Public Sector—2003–2018. \*Includes investment of the federal public enterprises. *Source:* Brazilian Statistical Office: Quarterly National Accounts; Fiscal Policy Observatory (IBRE-FGV), available at: <https://observatorio-politica-fiscal.ibre.fgv.br/posts/investimentos-publicos-1947-2018>; <https://observatorio-politica-fiscal.ibre.fgv.br/posts/nova-serie-historica-de-investimentos-publicos-1947-2018>.

followed by capital accumulation.<sup>4</sup> Thus, the theoretical apparatus developed by Minsky, on corporate financing positions and endogenous financial fragility, provides the tools to establish the connection between financial integration of developing economies and its effects on both the capital accumulation and the economic stability of these economies. In this perspective, the behavior of the real exchange rate in addition to the real interest rate turns out to be the most important variables to guide investment decisions, as they capture the speculative movements of the flow of capital in developing economies.

Hence, the economic stability of financially integrated developing economies largely depends on the pressures exerted by external commitments on the balance sheet of both non-financial and financial firms. Actually, if on one hand, the abundance of liquidity in international financial markets creates the conditions for the expansion of domestic liquidity, on the other, the dependence on foreign lending reduces policy space as the economy becomes hostage to international events outside the domestic economy. Thus, volatile capital flows contribute to increasing the external fragility of developing economies,<sup>5</sup> as sudden changes in the expectations of international investors can result in a drastic reduction in the level of economic performance, through the contraction of aggregate demand and devaluation of the exchange rate.<sup>6</sup>

### Recent Brazilian economic performance and the investment cycle

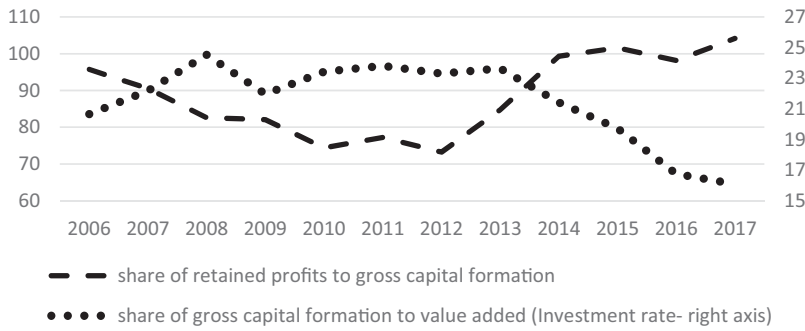
The recent investment cycle started in the first half of the 2000s when favorable external conditions stimulated the investment rate (Figure 1). This is also the period when the macroeconomic policy arrangement

(named as the macroeconomic policy tripod) based on the flex exchange rate, inflation targeting and primary fiscal surpluses was consolidated. During the commodities boom (2003–2010), the macroeconomic policy tripod was functional to keep inflation under control and to widen policy space for the implementation and expansion of social policies aiming at reducing income inequality and poverty.<sup>7</sup> Actually, growth during the 2000s rested on favorable terms of trade and on policies to boost domestic consumption.<sup>8</sup> However, the international financial crisis at the end of the decade negatively impacted the Brazilian economy, first through international credit crunch with consequences to the domestic credit market, and also through the decline in foreign trade.

According to Barbosa-Filho (2010, 1) “a sequence of unprecedented expansionary actions”<sup>9</sup> running from credit expansion to fiscal stimuli to boost aggregate demand and to avoid an excessive accumulation of inventories, especially of consumer durable goods, were well succeeded to recover the economy in 2010. In 2010, GDP growth rate was 7.5%, after a decline of 0.5% in 2009. But, as observed by de Paula, Modenesi, and Pires (2015), the economy started the 2010s in a more fragile situation and the European crisis in 2011–2012 imposed new challenges to the macroeconomic authorities. Therefore, the impact of the 2008 international financial crisis would already signal a reversal in the growth trajectory that started in 2003. The slow recovery of the world economy in the following years was reinforced by a change in domestic macroeconomic policy, which shifted from the incentive to aggregate demand to supply-side incentives instead, without recuperating growth rates. Therefore, the 2010s registers 2 years of deep recession (reduction of 3.6% in 2015 and 3.3% in 2016) with investment rates dropping steadily.<sup>10</sup>

A closer look at the evolution of the investment rate shows that it started to increase in 2004–2005 and gradually augmented until 2008. Public investment continued to grow until 2010, in an attempt by the government to sustain aggregate demand. The impact of the 2008 international financial crisis was felt in the investment rate in 2009, with a decrease of 2.1% in the growth rate of the gross fixed capital formation. The investment rate recuperated in 2010, when the gross fixed capital formation increased by 17.9% in that year, therefore, responding to the quick economic recovery. The investment rate was kept around 20.9% until 2013, but the domestic conditions to finance long-term investment and the entrepreneurial expectations were not the same as they were in 2008. In this sense, a sharp drop in the rate of investment is observed from 2013 onwards, followed by the drop in the public investment in 2014, due to changes in the macroeconomic policy arrangement.

In short, the expressive increase in the investment rate from the mid-2000s on was a result of the optimistic expectations in the potential growth



**Figure 2.** Non-financial firms: Share of retained profits to value-added and investment rate (%)—2006–2017. *Source:* Brazilian Statistical Office, National Accounts.

rate of the Brazilian economy directly related to the commodity boom prices. As a result of this favorable scenario, firms increased their indebtedness to finance productive investments, as expectations of future returns were that the expected cash flow would be sufficient to cover operating and financial costs. However, the sudden change in the external scenario in 2008 and the perspective of a long-lasting world recession, in line with Minsky's description of the business cycle, had a negative impact on business expectations, and the firms' financial balance sheet started to deteriorate.

Indeed, the first 2 years of the government of president Dilma Rousseff (2011–2012) was marked by a gradual worsening in the international scenario due to the Euro crisis and later by the decline in growth in emerging economies (including China). In order to deal with the new international environment, important changes in the structure and “modus operandi” of economic policy took place. According to Oreiro and D'Agostini (2017, 28), macroeconomic authorities tried to change the macroeconomic regime basically by making monetary policy more flexible (reduced interest rates and allowed inflation to reach the ceiling of the target range) and by reducing taxes on manufactured products.<sup>11</sup> This latter policy, a classical supply-side policy, aimed at increasing industrial competitiveness and was named the New Macroeconomic Matrix.<sup>12</sup> The new macroeconomic policy arrangement, however, did not succeed well, and according to Oreiro and D'Agostini (2017, 41), the failure is explained because it was unable to deal with the structural problems of the Brazilian economy, mainly low aggregate productivity and low competitiveness of manufacturing products. On the other hand, tax relief and other measures showed no clear effect to boost the economy due mostly to the deterioration of the agents' expectations. This deterioration can be explained because, according to de Paula, Modenesi, and Pires (2015), while the response to the 2008 international financial crisis involved a speedy series of substantial measures to relieve

taxes and expand spending, the response during the European crisis occurs only on the second half of 2012, and did not focus on sustaining public investments, with greater impact on aggregate demand.

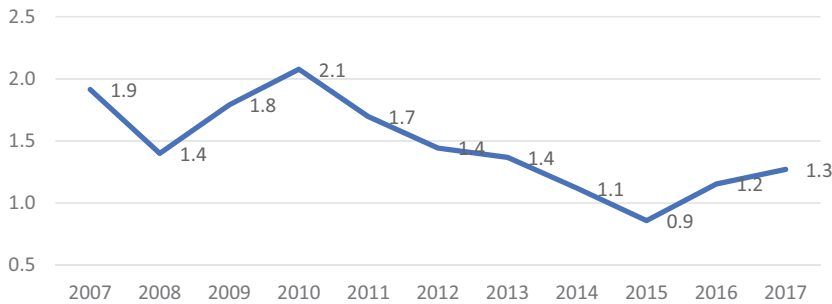
The impact of the deterioration in the expectations on firm's investment decision can be illustrated through the evolution of the share of non-financial firms retained profits in relation to capital formation (self-funding) and their investment rate (Figure 2). As long as the degree of confidence in expectations that guide investment decisions remained high, non-financial firms increased external funding. Indeed, in the ascending phase of the investment rate until 2008 and its resume in 2009 until 2011, non-financial firms' self-funding decreased indicating that they were increasing their indebtedness to fund their investment plans. However, once expectations became less optimistic, after 2012, non-financial firms changed their strategy and reduced investment rate while increasing profit retention. It should be added that the basic interest rate started to increase in 2012, moving from 7.25% per year to 14.25% per year in 2016.

The increase in the leverage ratio during the ascending phase of the investment rate was also followed by a change in the debt composition of non-financial firms. This change is characterized by an increase in the weight of foreign debt. Corrêa, Lemos, and Feijo (2017, 1142) show that, according to the Brazilian Central Bank, from the second quarter of 2007 to the second quarter of 2016, gross external debt of non-financial firms grew 142.6%.<sup>13</sup> Increasing indebtedness in foreign currency increases the foreign exchange exposure of firms and their sensitivity to international liquidity shocks.<sup>14</sup> Actually, the real exchange rate devalued by 24% from 2013 to 2015, the first year of depression.<sup>15</sup>

In this sense, we interpret that the Brazilian economy, due to the deterioration of expectations, started from 2012 to 2013 onwards the descending phase of a "Minskyan business cycle." Moreover, the economic measures taken in 2011 onwards to boost private investment showed little efficacy and shook confidence in the New Macroeconomic Matrix policy arrangement because they did not reverse business expectations.<sup>16</sup> The domestic macroeconomic indicators (public debt and inflation) were worse off, contributing to blurring the future prospects of the economy. From 2014 on, the Brazilian economy clearly entered a recessive phase.

### **An econometric exercise to explain financial fragility in the industrial sector: 2007–2017**

As seen in the previous section, the impact of the 2008 international financial crisis and the 2011–2012 Euro crisis and also the macroeconomic policy arrangement named as the New Macroeconomic Matrix on business



**Figure 3.** Indicator proxy of the financial fragility of the manufacturing and mining industries—2007–2017. *Source:* Brazilian Statistical Office: Annual Industrial Survey (own elaboration).

expectations led to a deceleration in demand for investment after 2013. After the 2015–2016 recession, the Brazilian economy still is in a slow recovery. A consequence of the poor evaluation about the prospects of the economy is that aggregate demand is constrained, affecting the balance sheet of indebted firms. In this context, balance sheets of firms become more fragile and susceptible to small changes in the financing conditions either in the domestic market or in the foreign markets, delaying a robust recovery in the growth rates.

The financial instability hypothesis states that the economy will turn more fragile, the faster the number of firms with difficulties to roll over past debt commitments grow. These difficulties arise when firms are not able to generate enough cash flow for an extended period. Minsky (1986, 200) defines a concept of gross-profit as quasi-rents, as follows: “Quasi-rents are the difference between the total revenue from selling output produced with the aid of capital assets and out-of-pocket, running, or technically determined costs associated with producing output; they are a gross-profit concept.” Besides the flow of quasi-rents, firms should also be able to contract new debts and roll over old ones when needed to keep business running. In order to be able to do so, they should keep liquid assets in their portfolio. These would work as margins of safety to allow firms to engage in debt financing. Therefore, together with the ability to generate quasi-rents, the margins of safety are part of the argument to explain the financial instability hypothesis. In Minsky’s words (1986, 241) “Two margins of safety for a firm that finances investment externally are the liquid assets held in portfolios and the excess of the present value of the expected quasi-rents from the project over the full costs of its completion.”

To contribute to the understanding of the investment cycle of the Brazilian economy in the 2000s and 2010s, this section aims to present an econometric exercise to explain the determinants of the financial fragility of industrial firms during the 2007–2017 period. We use panel data from the Annual Industrial Survey,<sup>17</sup> which is the most comprehensive collection

of annual data for the industrial sector, to build an aggregate financial fragility index covering all registered firms with one or more employees within the year in the manufacturing and mining sectors.<sup>18</sup> The aggregate financial fragility index is used in a cointegration model to explain the coevolution between investment rate and financial fragility of industrial firms.

Figure 3 shows the indicator of the financial fragility for industrial firms (manufacturing and mining).<sup>19</sup> This indicator measures the proportion of gross profits to non-operational expenditures.<sup>20</sup> The interpretation of this indicator is that when it is close or below 1, it means that the gross surplus does not cover contractual commitments, meaning that the balance sheets of non-financial firms are more fragile. According to Figure 3, the industrial surplus had been falling to debt commitments since 2010, although in 2012, for a short period, the share of surplus stopped falling. From 2013 onwards industrial surplus generation came close to non-operating commitments. In 2015 the indicator was below one for the first time.

Our approach to the financial fragility of the industrial firms considers the sensitivity of the firm with respect to the negative impact due to the divergence between the cash flow and debt commitments. In this sense, in order to estimate the econometric equation for the financial fragility, we assume that: (i) an increase in the availability of internal funds to the firm will have a negative impact on the financial fragility and (ii) an increase in the availability of external funds will have a positive impact on the financial fragility.

In our model, the impact of a greater exposition of the balance sheet of the firms will be captured by defensive strategies to face the increased leverage. Also, we will consider that the financial fragility varies with the business cycle, given that, according to Minsky's theory, a greater leverage implies higher probability of failures, if the availability of internal and external funds does not cover debt contractual commitments because aggregate demand is not sustained to validate past expectations.

We chose the following variables to explain the behavior of the financial fragility of industrial firms:

- i. the share of wages on value-added, assuming as in Feijo, Lamônica, and Bastos (2015) that the regime of accumulation during the 2000s and 2010s was a "weak" wage-led regime;<sup>21</sup>
- ii. the ratio between financial receipts to total receipts (both (i) and (ii) are proxies for the variation of the availability of internal funds) and
- iii. the investment rate aiming to capture the changes in the financial fragility due to the business cycle, which is assumed to be correlated with the availability of external funds to the firm.

A rise in the share of wages on value-added should reduce the availability of internal funds, and therefore should have a negative impact on the proxy of financial fragility. The opposite is expected with the rise in the share of financial receipts on total receipts. The reason for that is because, according to Almeida et al. (2013, 15), there is evidence that large firms in Brazil have been increasing their level of indebtedness. Analyzing 104 balance sheets from 2004–2007, the authors found that “[...] large Brazilian firms also kept financial investments and current assets growing. This indicates that a portion of the corporations’ increasing capacity to generate profits and liquidity, in addition to reducing their level of indebtedness, also worked to increase financial investments and turnover, which increased due to the greater expansion of production and sales” [free translation]. As seen, this finding is in line with Minsky’s analysis of margins of safety.

The expected sign of the investment rate is positive because it captures the evolution of the aggregate demand. In the ascending phase of the business cycle, an increase in the industrial investment rate would impact positively the total investment rate and aggregate demand, validating positive past expectations on the return on investment decision. As long as profits are sustained to cover past debt commitments, the availability of external funds to expand business may continue on the assumption that positive expectations about the future of the economy prevail. On the descending phase of the cycle, an opposite movement is expected to occur, with a negative impact on the generation of internal funds to sustain past debt commitments.

Our estimation strategy takes into account the simultaneous nature of the proxy for financial fragility and investment, and because of that we treat this endogeneity through the two-stages least-squares method. We instrumentalize two variables: the investment rate and the share of financial receipts to total receipts. The investment rate is instrumentalized based on the assumption that investments are a function of the expected future return. The financial investment, which proxy is the share of financial receipts to total receipts, is instrumentalized by the interest rate, the exchange rate and the external income.

Equation (1) presents the logarithmic specification of the financial fragility equation without control for the implicit endogeneity in the empirical relation. It explains the unobserved effect of the individual heterogeneity of the industrial sectors on the financial fragility of firms.

$$\ln F_{it} = \alpha_1 \ln S_{it-1} + \alpha_2 \ln I_{it-1} + \alpha_3 \ln R_{it-1} + \mu_i + \gamma_t + u_{it} \quad (1)$$

Where subscript (*i*) stands for sectors of activity and (*t*) for year and all variables are in log;

( $F_{it}$ ) is the proxy for the financial fragility of firms in the current time;  
 ( $S_{it-1}$ ) is the share of wages in relation to the value added lagged in 1 year;  
 ( $I_{it-1}$ ) is the share of the investment in fixed assets over the value-added lagged in 1 year; and  
 ( $R_{it-1}$ ) is the share of financial revenue in relation to the total revenue of firms lagged in 1 year. The terms  $\mu_i$  and  $\gamma_t$  are respectively the unobserved effect of the sectors and the temporal effect on the financial fragility of the firms; the  $u_{it}$  corresponds to the idiosyncratic error due to the individual and temporal effects of the sectors.

Equation (2) shows the proxy for the financial fragility of industrial firms controlling the implicit endogeneity of its determinants, that is to say, assuming that the  $E(u_{it}/R_{it-1}I_{it-1}) \neq 0$ .

$$\ln f_{it} = \varphi_1 \ln S_{it-1} + \varphi_2 \ln \bar{I}_{it-1} + \varphi_3 \ln \bar{R}_{it-1} + \mu_i + \gamma_t + \vartheta_{it} \quad (2)$$

Thus, its specification differs from Equation (1) by inserting the investment rate ( $\bar{I}_{it-1}$ ) and the ratio of financial receipts to total receipts ( $\bar{R}_{it-1}$ ) instrumented by the following variables:

( $E_t^e$ ) the expected commercial exchange rate in the end of the period, available at the Brazilian Central Bank;  
 ( $i_t^e$ ) the expected basic interest rate SELIC, available at the Brazilian Central Bank;  
 ( $Y^{*e}_t$ ) the expected world income, measured as the difference between the Brazilian GDP and the World GDP at constant price, available at the World Development Indicators of the World Bank and  
 ( $V_{it}^e$ ) the expected value added as a proxy of investment returns.

Considering that the specification of expectations is determined by past information, it is assumed that all instruments are the function of their respective values lagged in one period. The instrumentalized variables capture the effects of macroeconomic variables on the financial fragility.<sup>22</sup>

Equations (3) and (4) illustrate the specifications of the variables used in the financial fragility equation. Instrumental variable of investment according to the instrument of expectation of value-added:

$$V_{it}^e \quad \text{with} \quad V_{it}^e = \theta_4 V^{\omega_1}_{it-1} \quad (3)$$

Instrumental variable of the share of the financial receipt as a function of the instruments expectation of the exchange rate ( $E_t^e$ ), interest rate ( $i_t^e$ ), and world income ( $Y^{*e}_t$ ).

**Table 1.** Panel unit root tests and integration order (d).

| Tests         | IPS        | I(d) | ADF         | I (d) | PP          | I(d) | HLM       | I (d) |
|---------------|------------|------|-------------|-------|-------------|------|-----------|-------|
| Inf           | -2.5149*** | 0    | 82.5842***  | 0     | 82.5842***  | 0    | 4.5663*** | 1     |
| $\Delta$ Inf  | -4.0048*** |      | 287.5448*** |       | 287.5448*** |      | -1.4194   |       |
| lnS           | -2.6051*** | 0    | 83.2742***  | 0     | 83.2742***  | 0    | 5.4179*** | 1     |
| $\Delta$ lnS  | -3.9148*** |      | 289.9710*** |       | 289.9710*** |      | -2.0778   |       |
| lnI           | -2.9378*** | 0    | 134.5478*** | 0     | 134.5478*** | 0    | 3.3802*** | 1     |
| $\Delta$ lnI  | -4.5846    |      | 403.0844*** |       | 403.0844*** |      | -2.9463   |       |
| lnR           | -2.1045*** | 0    | 38.3384     | 1     | 38.3384     | 1    | 3.3186*** | 1     |
| $\Delta$ lnR  | -2.5282*** |      | 77.6241***  |       | 77.6241     |      | -0.6096   |       |
| lni           | -2.0183*** | 0    | 26.2599     | 2     | 26.2599     | 2    | 5.0458*** | 1     |
| $\Delta$ lni  | -2.0275*** |      | 70.7036**   |       | 70.7036**   |      | -0.1018   |       |
| lnE           | -1.9640*** | 0    | 23.8275     | 1     | 23.8275     | 1    | 6.8199*** | 1     |
| $\Delta$ lnE  | -3.6766*** |      | 186.6007*** |       | 186.6007*** |      | -1.2354   |       |
| lnY*          | -2.5558*** | 0    | 60.0932     | 1     | 60.0932     | 1    | 7.3958*** | 1     |
| $\Delta$ lnY* | -3.5155*** |      | 163.8257*** |       | 163.8257*** |      | -1.7609   |       |
| lnV           | -2.2812*** | 0    | 67.2546     | 1     | 67.2546     | 1    | 8.8210*** | 1     |
| $\Delta$ lnV  | -3.3300*** |      | 186.1907*** |       | 186.1907*** |      | -2.0426   |       |

Note: Hypothesis: IPS, ADF, PP. Ho: All panels contain unit roots; Ha: At least one panel is stationary or Some panels are stationary; Hypothesis: HLM; Ho: All panels are stationary; Ha: Some panels contain unit roots. \*\*\*Significance level: 1%.

$$\bar{R}_{it} = E_t^e i_t^e Y^{*e}_t \quad (4)$$

Where  $E_t^e = \theta_1 E^{\beta_1}_{t-1}$ ;  $i_t^e = \theta_2 i^{\beta_2}_{t-1}$ ;  $e Y^{*e}_t = \theta_3 Y^{*\beta_3}_{t-1}$

The most rigorous assessment of the long-term relationship between the investment rate and financial fragility is carried out through the panel's cointegration analysis. Table 1 shows the order of integration of the model variables through several unit root tests. The order of integration is given by the number of differences necessary to make the series stationary. In Table 1, tests were performed on the variables at level and first difference, given by the difference operator  $\Delta$ . The IPS tests suggest that some panels are stationary in level in all variables, therefore, integrated in order zero I (0). The ADF and PP tests arrive at identical results, accept the hypothesis of some stationary panels I (0) for financial fragility, wage rate and investment rate; I (1) for the share of financial revenue in total revenue, exchange rate, foreign income and added value; and I (2) for interest rates. However, there is an inaccuracy in the hypothesis of a stationary panel of these tests, as they accept that there are some stationary panels. The HLM test is the only one that accepts the null hypothesis that all panels are stationary in the first difference. According to this test, all variables are I (1), where in the first difference the hypothesis that all panels are stationary is not rejected.

The ADF, PP and PPM statistics of the Pedroni integration test reject the hypothesis of absence of cointegration at 1% of significance, with one and two lags, considering financial fragility as a dependent variable (Table 2). According to the tests, there is a long-term stability relationship between the model variables. The results suggest that, even under the conditions of high interest rates, exchange rate fluctuation, and the

**Table 2.** Pedroni cointegration test.

| Tests | Lags   | Panel <sup>a</sup> | Panel <sup>b</sup> |
|-------|--------|--------------------|--------------------|
| ADF   | 1      | −12.7500***        | −6.4014***         |
|       | 2      | −10.9753***        | −15.4038***        |
| PP    | 1 to 2 | 7.0048***          | −7.1999***         |
| PPM   |        | −11.1861***        | 9.4629***          |

Note: Augmented Dickey–Fuller (ADF); Phillips–Perron (PP); Modified Phillips–Perron (PPM).

Hypothesis Pedroni test:  $H_0$  is No cointegration and  $H_a$  are All panels cointegrated.

<sup>a</sup>Alternative hypothesis: AR parameter (within dimension).

<sup>b</sup>Alternative hypothesis: AR parameter (between dimension).

\*\*\*Significance level: 1%.

**Table 3.** First stage equations of the estimator with endogeneity control.

| Variables                                             | $(I_{t-1})$ Investment $_{t-1}$ | $(R_{t-1})$ Financial receipt $_{t-1}$ |
|-------------------------------------------------------|---------------------------------|----------------------------------------|
| $E_{t-2}$                                             | −0.5634424*** (0.1802919)       | 1.162329*** (0.208373)                 |
| $i_{t-2}$                                             | 0.188311 (0.1549501)            | 0.564052*** (0.1758703)                |
| $Y^*_{t-2}$                                           | −0.3289942 (0.7541713)          | −2.234458*** (0.849075)                |
| $V_{it-2}$                                            | −0.2130728 (0.3450729)          | 0.9677707* (0.5553746)                 |
| $S_{it-1}$                                            | 1.179628*** (0.230467)          | 0.4447243 (0.3057048)                  |
| Statistical summary of the equations on the 1st stage |                                 |                                        |
| Test $F$                                              | 10.03                           | 33.21                                  |
| Prob $F$                                              | 0.0000                          | 0.0000                                 |
| SWM $F$                                               | 7.56                            | 11.83                                  |
| Prob $F$                                              | 0.0001                          | 0.0000                                 |
| SW $\chi^2$                                           | 23.27                           | 36.42                                  |
| Prob $\chi^2$                                         | 0.0000                          | 0.0000                                 |
| SW $F$                                                | 10.03                           | 33.21                                  |
| Prob $F$                                              | 0.0000                          | 0.0000                                 |
| Anderson–Rubin $F$                                    |                                 | 19.14                                  |
| Prob $F$                                              |                                 | 0.0000                                 |
| Anderson–Rubin $\chi^2$                               |                                 | 78.55                                  |
| Prob $\chi^2$                                         |                                 | 0.0000                                 |
| Stock–Wright $\chi^2$                                 |                                 | 38.54                                  |
| Prob $\chi^2$                                         |                                 | 0.0000                                 |
| Chow $F$                                              |                                 | 1.97                                   |
| Prob $F$                                              |                                 | 0.0066                                 |
| Hausman test $\chi^2$                                 |                                 | 22.61                                  |
| Prob $\chi^2$                                         |                                 | 0.0000                                 |

Note: \*\*\* 1% significance; \* 10%; between brackets are the standard deviation estimates of the parameters.

deceleration of the world economy, there is a balance between the investment strategies in capital formation and the share of liquid assets in portfolio. Thus, given the conditions of order of integration and stability, it becomes possible to estimate the empirical equation of financial fragility with fixed effect in two stages, presented in [Tables 3](#) and [4](#).

The two-stage estimation of [equation 2](#) presents the investment and financial receipts with instrumental variables in the first stage ([Table 3](#)). [Table 4](#) presents the financial fragility equation in the second stage with the variables already instrumentalized. The estimator with random and pooled effects are rejected in favor of the fixed effect estimator, as suggested by the Hausman and Chow tests in [Table 3](#), significant at 1%. The fixed effect estimator suggests that individual heterogeneity, not observed, is also relevant to explain financial fragility.

**Table 4.** Second stage equations of the financial fragility model.

| Variables                       | ( $f_i$ ) Financial Fragility Index |
|---------------------------------|-------------------------------------|
| $S_{it-1}$                      | -2.724201*** (0.445605)             |
| $I_{it-1}$                      | 1.314267*** (0.375679)              |
| $R_{it-1}$                      | 0.7197465*** (0.1966791)            |
| Kleibergen–Paap $\chi^2$        | 18.771                              |
| Prob $\chi^2$                   | 0.0003                              |
| Hansen $\chi^2$                 | 0.010                               |
| Prob $\chi^2$                   | 0.9949                              |
| Teste de endogeneidade $\chi^2$ | 35.510                              |
| Prob $\chi^2$                   | 0.0000                              |

Note: \*\*\* 1% significance; between brackets are the standard deviation estimates of the parameters.

In the first stage, there is no statistically significant effect of the expected return on value added ( $V_{it-2}$ ) on the investment rate. However, the instrumentalization of the investment rate shows two interesting results in relation to exchange rate shocks and labor costs, with statistical significance of 1%. As the lagged exchange rate ( $E_{it-2}$ ) is negatively affecting investment, it shows that exchange devaluation are making investments more expensive. In relation to increase share of wages ( $S_{it-1}$ ), the positive effect on investments reinforces the theory of the marginal factor replacement rate, caused by exogenous shocks in the relative prices of factors of production. However, it is the instrumentalization of financial receipts that shows the most relevant results, by demonstrating the positive effect of exchange rate devaluation and the domestic interest rate on the financial receipts of firms, at 1% significance. Also, the slowdown in the world economy affects at 1% significance the share of financial receipts, indicating that firms seek for safety to preserve their income.

The tests of the second stage equation (Table 4) confirm the statistical relevance of the instruments.<sup>23</sup> The results show that the effect of lagged investment rate ( $I_{it-1}$ ) is greater than the effect of lagged share of financial receipts to total receipts ( $R_{it-1}$ ) on financial fragility. Thus, it is concluded that financial fragility tends to increase as the pace of investment decreases. For every 1% reduction in the investment rate, there is a decrease of approximately 1.31% in the financial fragility indicator, compared to the 0.71% increase for each 1% increase in the increase in the share of financial receipt to total receipt. The greater effect of lagged rate of investment ( $I_{it-1}$ ) on financial fragility confirms the importance of the phases of the business cycle on building the financial fragility of firms. Even so, the variable with the greatest elasticity over financial fragility is the lagged wage rate ( $S_{it-1}$ ). The 1% increase in wages reduces the financial fragility index by approximately 2.72%.

In sum, the results in Table 4 corroborate the hypothesis that a context of deceleration of the aggregate demand and high uncertainty amplifies the divergence between the generation of internal funds and contractual commitments, increasing the financial fragility of firms. The steady decline of

the financial fragility index of industrial firms (Figure 3) after 2010 indicates that the ascending phase of the investment cycle was coming close to a turning point. Indeed, the European crisis in 2011–2012 negatively impacted the Brazilian economy, turning expectations of foreign investors more volatile. In our model, the increasing external fragility of the Brazilian economy was captured through the effect of the exchange rate on the instrumentalization of two of the explanatory variables—the investment rate and the share of the financial receipts to total receipts.

Besides the increased external instability, the new economic policy arrangement starting in 2011, combining the macroeconomic tripod with supply-side incentives to the private sector, proved to be a poor arrangement to countervail the exacerbation in business uncertainty and deceleration of aggregate demand. Indeed, the most important variable to explain the financial fragility of the industrial firms was the wage share. The negative impact of this variable in the financial fragility points out to the inefficacy of the program of tax reductions designed to reduce payroll costs of industrial firms, a supply-side measure, instead of a public policy aiming at boosting public investment. According to our model, it is the investment rate the variable that most explains the reduction of the financial fragility.

### Concluding remark

The recent Brazilian economic crisis, different from past experiences, has not been caused by an international liquidity crisis; domestic factors, such as inconsistent political economy, together with a fragile world economy recovery since 2008, largely explain the 2015–2016 recession and the poor economic performance since then. We have argued in this paper that the 2008 international financial crisis hit the Brazilian economy in an ascending phase of a typical Minskyan cycle, and therefore, the main cause of the sluggish recovery is due to the low rate of growth of aggregate demand and the high level of financial fragility of non-financial firms. In our view, the only way out of the scenario of low growth is through the resumption of public and private investment. However, as argued, to resume private investment it is necessary to improve the balance sheet of firms, and the recovery of sound financial indicators requires a scenario of sustained growth. This is a particularly difficult task to be accomplished considering that the asymmetrical financial insertion of the Brazilian economy reduces its policy space.

### Notes

1. In brief, Minsky (1992, 7) says: “The financial instability hypothesis, therefore, is a theory of the impact of debt on system behavior and also incorporates the manner in which debt is validated.”

2. The role of banks and other financial institutions is discussed, for instance, in Carvalho (2016).
3. On the asymmetrical hierarchy of currencies, see de Paula, Fritz, and Prates (2017).
4. Kregel (2004, 580) when analysing the position of developing economies with net international lending concludes that “by definition and by design, [developing economies will be] operating a “Ponzi” financial profile since the interest and amortization on the borrowing can only be met by additional foreign capital inflows.” This implies that for a country to be successful in engaging in foreign indebtedness to finance its growth in a financially integrated world means that “its success depends on the willingness of foreign lenders to continue to lend.”
5. See, for instance, Karwowski and Stockhammer (2017).
6. The concept of ‘balance of payments dominance’, as proposed by Ocampo (2013), offers an explanation for the external fragility of these economies, highlighting the importance of capital account in causing cyclical shocks. Financial markets, under the assumption of non-probabilistic uncertainty, adopt practices and procedures leading to risk evaluation, which tend to stress the financial cycle derived from the capital account of financially integrated developing economies. In this context, macroeconomic policies in developing countries tend to be procyclical. There are three main consequences to the macroeconomic policy management stemming from the dependence of financially integrated developing economies. One is that nominal and real interest rates tend to be relatively higher; the second is that real exchange rates tend to be appreciated over time. The third is that since both trends can lead these economies to reduce their long-term capacity to grow, fiscal policy is procyclical, limiting the space for growth policies. For a discussion on the constraints to grow with foreign savings in the Brazilian economy, see, for instance, Bresser-Pereira and Nakano (2003); Feijo, Lamônica, and Bastos (2016) and Feijo and Lamônica (2019).
7. Rey (2015, abstract), in influential research, concludes that “There is a global financial cycle in capital flows, asset prices and in credit growth .... Assets markets in countries with more credit inflows are more sensitive to the global cycle.” The author shows that no matter the exchange rate regime, economies financially integrated face a ‘dilemma’ of political economy and not a ‘trilemma’ as in the Mundell-Fleming model. In the absence of capital controls, monetary policy loses autonomy, mainly in economies dependent on foreign savings.
8. de Paula, Modenesi, and Pires (2015) refer to the Brazilian growth in 2004–2008 as a miniboom.
9. Domestic demand was fueled by several stimuli, such as: the improvement in income distribution (through the increase of the real minimum wage and income policies to fight inequality); an increase in employment rates and an increase in the supply of consumer credit.
10. For a list of the measures, see de Paula, Modenesi, and Pires (2015).
11. The liberal economic reforms that have been implemented since 2016, as a way out of the deep recession, have not yet delivered the resumption of growth, as they have not reversed expectations in favor of a sustained recovery. It should also be remarked that since 2016, the fight against corruption in the country, through the operation named ‘lava-jato’ has also contributed to the increased uncertainty in the economy.
12. Tax reductions were mainly designed to reduce corporate costs such as cuts in payroll.
13. Ferrari-Filho, Cunha, and Bichara (2014, 536) point out that here was an inconsistency between the Keynesian approach of New Macroeconomic Matrix and the macroeconomia tripod based on the New Macroeconomic Consensus.

14. The authors also quote the CEMEC (2016) study that shows that among the joint-stock firms, excluding Petrobras, the share of debt in foreign currency increased from 24.2% in 2010 to 46.3% in the second quarter of 2016. Including Petrobras, the situation is more alarming, since foreign currency debt increased from 32.7% in 2010 to 60.0% in the second half of 2016.
15. As for the foreign indebtedness of Latin American firms, Pérez-Caldentey, Favreau-Negront, and Lobos (2018) show that the composition of the external debt in Latin America and Caribbean has changed significantly during the 2007–2015 period, although the level of the external debt in relation to GDP had been rather stable, around 18% of GDP of the region. According to the authors, the debt of non-financial firms raised from 14.8% in total debt in 2007 to 32.9% in 2017. Based on a sample of 2,241 non-financial firms in Latin America (1,281 Brazilian firms) with open capital, the authors also find that during the 2007–2015 the financial fragility of firms with debts in foreign currency had increased significantly.
16. The index of real exchange rate (average of the year) was 89.89 in 2013 and 111.58 in 2015 (serie 11752 from Brazilian Central Bank).
17. See Serrano and Summa (2015); Arestis, Baltar, and Prates (2015) and Corrêa, Lemos, and Feijo (2017).
18. For an alternative proxy for financial fragility after the 2008 international financial crisis, with a sample of large firms, see Almeida, Novais, and Rocha (2016).
19. The Annual Industrial Survey is produced by the Brazilian Statistical Office (IBGE). For information on the methodology and sampling procedures, see: <https://www.ibge.gov.br/estatisticas-novportal/economicas/industria/9042-pesquisa-industrial-anual.html?=&t=conceitos-e-metodos>; Accessed March 10, 2020, January 29, 2019.
20. Industrial investment accounts for most of the investment in machinery and equipment in the Gross Capital Formation.
21. This indicator compares cash flow after of-pocket- costs and taxes with current receipts. It is built as the ratio between total receipts minus total costs and expenditures, excluding taxes and duties, depreciation and non-operational expenditures, divided by the sum of taxes and duties, depreciation end non-operational expenditures. Data are from the Annual Industrial Survey, IBGE.
22. The authors, based on an econometric study about the degree of indebtedness of non-financial firms (op.cit, 14), conclude that: “One insight of our paper is that a wage-led demand regime would hardly imply an increase in capital accumulation in a scenario of persistent real exchange rate overvaluation. On the opposite, exchange rate misalignment increased wage costs, reducing industrial competitiveness, negatively affecting the growth rate of the economy. Also, it contributed to substitute domestic for imported goods. Therefore, we can say the Brazilian economy is in a ‘weak’ wage-led demand regime, in the sense that it has not been able to sustain high growth rates, mostly after 2010.”
23. The tests performed to evaluate the statistical relevance of instrumental variables are of three types: subidentification, overidentification and extraneous exogeneity. Subidentification tests are performed on first- and second-stage equations with the null hypothesis that endogenous regressors are not relevant or are poorly identified. The over-identification test is performed only in the second stage of the equation, and has as null hypothesis the restriction of excessive identification. The strict exogeneity hypothesis is a test of the second stage equation, this test is determinant in the choice of an endogenous modeling.
24. The tests of the first stage equation show the statistical relevance of the instruments, as can be seen from agreements with the tests listed in the statistical summary of the first

stage equations. The Sanderson–Windmeijer (SW) statistics reject the hypothesis of underidentification and weak identification of individual endogenous regressors, with a 1% probability. Likewise, the Anderson–Rubin and Stock–Wright tests, robust to the presence of weak instruments, reject the joint hypothesis of endogenous regressors equal to zero, also at 1% significance. The second stage tests reinforce the results of the first stage on the relevance of the instruments. The endogeneity test rejects the hypothesis of extreme exogeneity of the endogenous regressors at 1% significance. The Kleibergen–Paap underidentification tests in the second stage, reject the hypothesis that the excluded instruments are not relevant to 1%, accepting that the model is identified, and the Hansen overidentification test accepts the hypothesis that the instruments identify the model. Therefore, there are no restrictions on the instruments used.

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